

Session 1 homework — MILK10k exploratory data analysis

Computer Vision and Speech Recognition · Stanislaus Lattorff

GitHub repository

Public repository (my personal account):

<https://github.com/stansilaisijfneq/milk10k-computer-vision>

The EDA notebook is at `notebooks/01_eda_milk10k.ipynb`. It is saved with all the outputs so the plots are visible on GitHub without running anything. The code it uses is in `src/milk10k/` and the figures it produces are in `reports/figures/`. The dataset is not in the repo (345 MB and CC BY-NC), so there is a script `scripts/download_data.sh` that downloads it from ISIC.

The clinical task

MILK10k contains photos of skin lesions from patients who were sent to a dermatologist. Every lesion was photographed twice: once as a normal close-up photo, and once through a dermatoscope, which is a lens with polarised light that makes structures below the skin surface visible. Almost all of these lesions were then biopsied, so the labels come from a pathologist looking at the tissue under a microscope rather than from someone's opinion.

The task is to predict from the images whether a lesion is benign, malignant or indeterminate (the `diagnosis_1` column). There is also a more detailed 11-class version of the label as an optional target.

Why this matters is mostly about timing. A skin cancer that is found early is usually removed in a small operation and that is the end of it. The same cancer found late is a much bigger problem, and for melanoma it can be fatal. So the model is a triage tool. It should almost never call something benign when it is actually cancer. Missing a melanoma is much worse than doing one unnecessary biopsy, so the number I care about is recall on the malignant class, not accuracy.

One thing I think is important to say clearly: this dataset only contains lesions that a doctor already found suspicious enough to biopsy. 69% of them are malignant, which is nothing like the general population, where almost everything you look at is a harmless mole. So a model trained on this data is really answering "given that a specialist was already worried about this lesion, is it cancer?" and not "is this cancer?". It would not work as a phone app for the public. A longer version of this is in `docs/clinical_task.md` in the repo.

What I did

- Downloaded the full dataset (10,480 images, 5,240 lesions) and loaded `metadata.csv` and `training_gt.csv` with pandas.
- Checked the structure rather than assuming it. Every lesion does have exactly two images, one of each type, and the 11-class labels are single-label.
- Plotted the class distribution for both targets, the missing values, age, sex, body site, skin tone, and the image sizes and colour statistics.

- Looked at the actual images: one example per class, and pairs of the same lesion in both modalities.

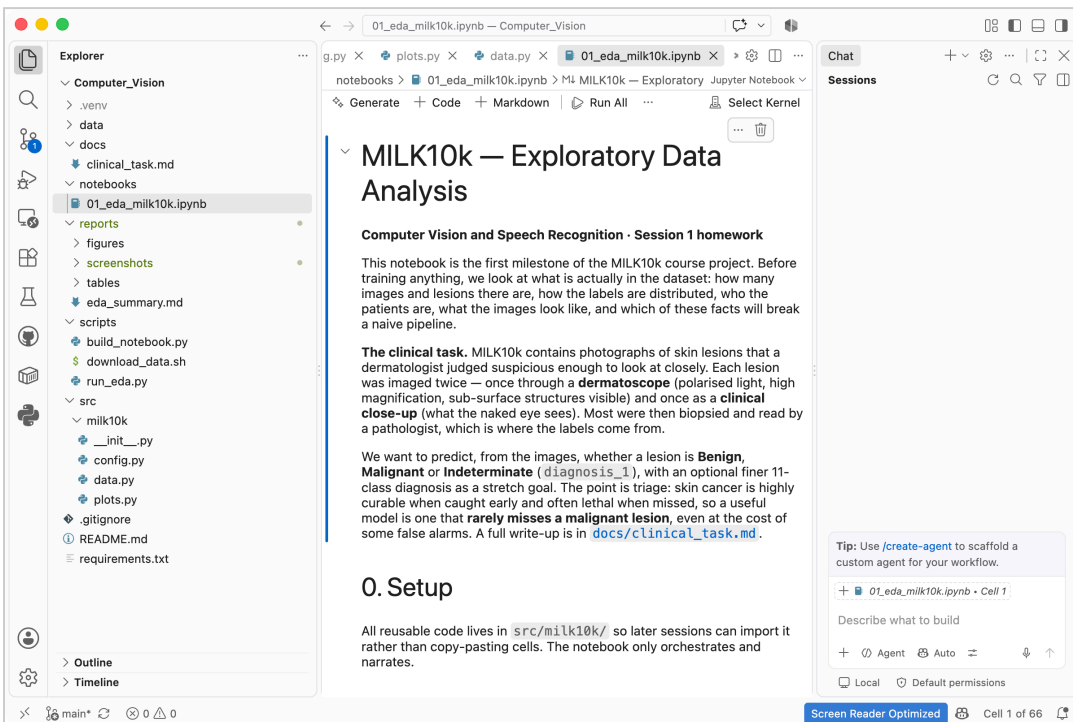
A few things I noticed

- The dataset is 69.4% malignant. Predicting "malignant" for everything already gives 69.4% accuracy, so accuracy on its own does not tell you anything here.
- Since every lesion has two images, a random split by row can put both images of the same lesion on opposite sides of train and test. Then the model has basically already seen the test case. The split has to group by `lesion_id`.
- The 11-class target is very unbalanced. BCC has 2,522 lesions and MAL_OTH has 9.
- AKIEC does not map to one single value of `diagnosis_1`. It is split between Indeterminate (123) and Malignant (180), so I cannot just translate the 11-class label into the 3-class one with a lookup table.
- 61% of the lesions are one skin tone class. If I only report the overall score I will not see how the model does on the tones that are rare in the data.

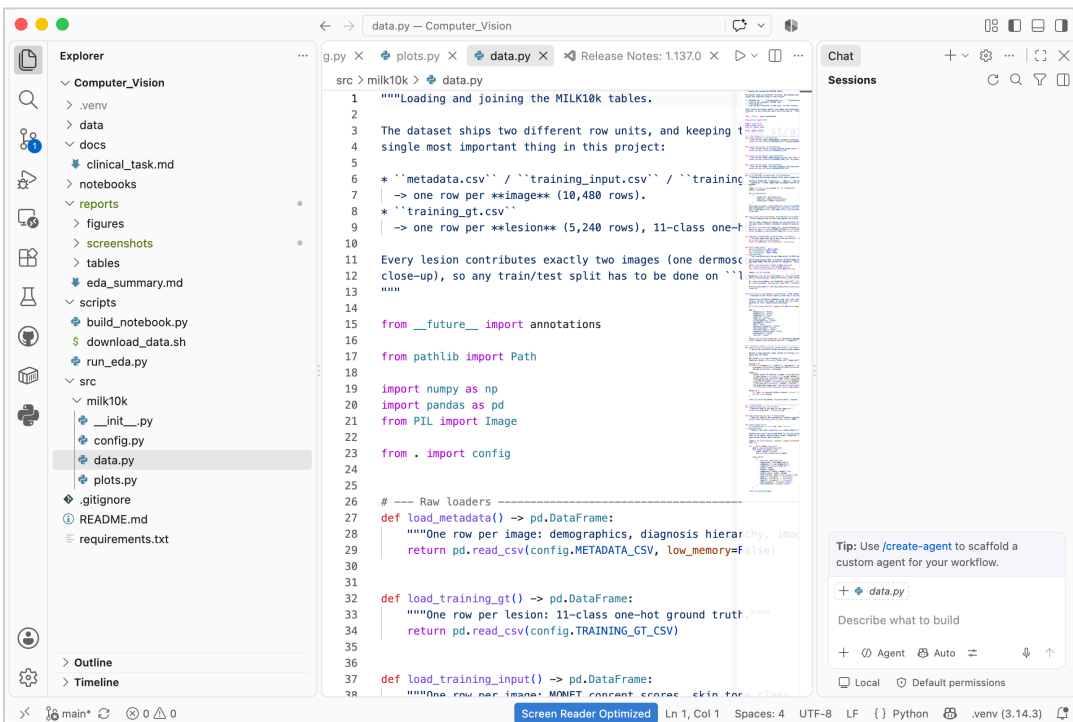
Editor

I am using Visual Studio Code. Screenshots of the project on the next page.

Code editor with the project structure



The project in VS Code. Folder structure on the left, the EDA notebook open.



The same project with `src/milk10k/data.py` open, which is where the loading and the structure checks are.